Internal exchange and controlled variational coexistence

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This is one newly hypothesized parent, not a repair of P1 or PAH. The immutable contract is TECT-EXCH-001-v1, SHA-256 `d07cb738d40d9b97a3a2341ada94ac13167e4410edd4ece39fb6ba02504f073e`. We prove genuine component exchange and a controlled shifted finite-volume variational coexistence. We separately reject exact inheritance of the old R-158 coexistence point. The host C6 remains T1 OPEN and T-054 unchanged.

On the original periodic torus of side 16 and volume $V = 4096$, fix any finite integer $K \geq 1$. The three-component Ψ has Fourier indices $|n_i| \leq K$; the scalar χ has $|n_i| \leq 2K$. Use exact integrals, orthonormal Fourier coordinates and orthogonal variational projections, not collocation products. All original P1 coefficients and positive floors remain. The new dimensionless inputs are $\omega = u = 1$, $\varepsilon = 1/16$. Writing $\rho = |\Psi|^2$ and $S = \sum_a \Psi_a^2$, set

$$H = F_{\mathrm{P1}}[\Psi] + \int \left(\frac{|\chi|^2}{2} + \frac{|\chi|^4}{4} - \varepsilon \Re(\bar{\chi} S) \right), \quad Q = Q_\Psi + 2Q_\chi, \quad Q_\Psi = \frac{1}{2}\int \rho, \quad Q_\chi = \frac{1}{2}\int |\chi|^2.$$

The real canonical bracket is $\{z, \bar{z}\} = -2i$ in each coordinate; $\dot{z} = -2i\partial_{\bar{z}} H$ uses a new parameter τ. It is not A2 gradient time, PAH Markov time or an observed clock. No Gibbs law or quantum state is assumed. Static states mean all minimizers of $\Omega_\mu = H - \mu Q$. Here μ is a multiplier, not a fitted forcing history or a predicted cosmological control.

2 Inherited input and exact dynamics

The conditional R-158 identity, restricted to this Fourier subspace, is

$$F_{\mathrm{P1}} = \mu_t Q_\Psi + \tfrac{1}{2}\langle \Psi, (L - \lambda_0)\Psi \rangle + F_{\mathrm{II}} + \frac{\gamma}{6}\int \rho(\rho - \rho_*)^2.$$

All displayed remainders are nonnegative. Its source provides $\gamma = 81/50$, $\rho_* = 43/216$, $g = \lambda_0 - \mu_t = 1849/86400$ and $1/4 < \mu_t < 1/3$. These are inherited P1 inputs, not new physical constants. The real unit lowest internal eigenvector v_* and shell $k = (2\pi/16)(1,1,1)$ give $\Psi_* = \sqrt{\rho_*} v_* e^{ikx}$. Then $S_* = \rho_* e^{2ikx}$ and $F_{\mathrm{II}}[\Psi_*] = 0$.

Global phase invariance of the unchanged P1 term follows from its quadratic, density and bilinear-current definitions; it is not local gauge invariance. The full parent has the weighted symmetry $(\Psi, \chi) \mapsto (e^{i\theta}\Psi, e^{2i\theta}\chi)$. Direct real-coordinate differentiation yields

$$\dot{Q}_\Psi = 2\varepsilon I, \qquad \dot{Q}_\chi = -\varepsilon I, \qquad I = \Im \int \bar{\chi} S, \qquad \dot{Q} = 0.$$

Orthogonal Fourier projection preserves these integrated identities. For constant $\Psi = (1,0,0)$ and $\chi = -i$, $\dot{Q}_\Psi / V = 2\varepsilon \neq 0$. The weights are essential: the unweighted sum is not conserved. At fixed finite K,

positive floors make the Hamiltonian vector field smooth. Each level of the positive quadratic Q is compact, so local ODE solutions cannot escape in finite positive or negative time. They extend to a complete finite-dimensional Hamiltonian flow. This is no infinite-dimensional flow theorem.

3 Exact old coexistence is not inherited

At $(\Psi_*, 0)$ the χ derivative is nonzero. Its doubled shell is admitted by the preregistered cutoff. At μ_t take $\chi = ce^{2ikx}$ with $c = \varepsilon\rho_*$. The change per volume is

$$\frac{\Omega_{\mu_t}(\Psi_*, \chi)}{V} = -(1/2 + \mu_t)c^2 + c^4/4 < 0, \qquad 0 < c < 1.$$

But $\Omega_{\mu_t}(0, 0) = 0$. Thus the old coexistence value and embedded nonzero minimizer do not survive unchanged. This neither refutes R-158 on its own functional nor rules out a shifted transition of this parent.

4 Controlled shifted coexistence

For $\mu \le \mu_t$, $a = 1/2 - \mu \ge a_0 = 1/6$. Pointwise $|S| \le \rho$ and square completion give

$$a|\chi|^2 - \varepsilon\Re(\bar{\chi}S) \ge -\frac{\varepsilon^2|S|^2}{4a} \ge -\frac{\varepsilon^2\rho^2}{4a_0}.$$

Put $A = \gamma/6$, $b = \varepsilon^2/(4a_0)$ and

$$D = 2b\rho_* + \frac{b^2}{2A} = \frac{2827}{1179648}, \qquad \mu_- = \mu_t - 2D > 0.$$

For $\delta = \mu_t - \mu$, the density bracket has the exact form

$$\frac{\delta}{2} + A(\rho - \rho_*)^2 - b\rho = A\left(\rho - \rho_* - \frac{b}{2A}\right)^2 + \frac{\delta - D}{2}.$$

Hence at μ_-, $\Omega \ge DQ_\Psi$, and if $\Psi = 0$ the remaining $a|\chi|^2 + |\chi|^4/4$ is strictly positive unless $\chi = 0$. Zero is the unique minimizer at μ_- and below. The preceding legal trial has negative energy at μ_t.

Compactness, not a finite table. Minimizers exist uniformly over the compact μ interval at each fixed K. Indeed $F_{\mathrm{P1}} \ge \lambda_0 Q_\Psi + (\lambda/4)\int\rho^2 + (\gamma/6)\int\rho^3$, and $-\varepsilon|\chi|\rho \ge -\varepsilon(|\chi|^2 + \rho^2)/2$. Since $a_0 - \varepsilon/2 > 0$, the sextic Ψ and quartic χ terms dominate the remaining lower-degree polynomials. This is equicoercivity in the finite Fourier coordinate space, so sublevel sets are compact.

Explicit exclusion of a continuous collapse to zero. Let $M_\Psi = (2K+1)^3$, $M_\chi = (4K+1)^3$ and

$$c_0 = \min(g, a_0), \quad A_K = 2\varepsilon\sqrt{M_\chi/V}, \quad B_K = |\lambda|M_\Psi/V, \quad \eta_K = \min\left\{\left(\frac{c_0}{4A_K}\right)^2, \frac{c_0}{4B_K}\right\} > 0.$$

The exact Fourier bounds $\|f\|_\infty \le \sqrt{M/V}\|f\|_2$, $\|\Psi\|_2^2 \le 2Q$ and $\|\chi\|_2^2 \le Q$ imply

$$\Omega_\mu \ge c_0 Q - A_K Q^{3/2} - B_K Q^2 \ge \tfrac{1}{2}c_0 Q > 0 \quad (0 < Q \le \eta_K).$$

Here nonnegative sextic, reservoir quartic and Class-II terms were dropped, not replaced. Thus a sequence of negative-energy minimizers cannot collapse to zero, uniformly in μ on this interval, at this fixed K.

Threshold and jump. Let $m_K(\mu) = \min \Omega_\mu \leq 0$. Equicoercivity and continuity imply continuity of m_K; it is nonincreasing since $Q \geq 0$. The small-charge exclusion and compactness also show a neighborhood above μ_- still has no nonzero minimizer of nonpositive energy: otherwise compact minimizers would converge to a nonzero zero-energy state at μ_-. Define $\mu_c(K) = \inf\{\mu \in [\mu_-, \mu_t] : m_K(\mu) < 0\}$. It lies strictly between μ_- and μ_t. Approach it from above with minimizing states. A compact subsequence converges to a minimizer $z_c \neq 0$, with $Q(z_c) \geq \eta_K$, and $m_K(\mu_c) = 0$. Zero and z_c coexist, and $\Psi_c \neq 0$ because pure reservoir states have strictly positive energy here. Below μ_c any nonzero zero-energy minimizer would give negative energy at a slightly larger multiplier below μ_c, a contradiction. Above μ_c, the trial z_c gives $m_K(\mu) \leq -(\mu - \mu_c)Q(z_c) < 0$. This is a finite-regulator variational first-order onset, with

$$0 < \mu_t - \mu_c(K) < 2D = \frac{2827}{589824} < 0.004793.$$

The window is common to finite K, but η_K depends on K. No uniform jump, cutoff limit, thermodynamic transition or continuum dynamics has been obtained.

5 Devil's-advocate review and verification

1. **Half factors and conserved quantity.** Dismissed within the declared convention by independent real Hamilton equations and a nonzero constant-mode witness. Density-only coupling is a negative control with no transfer.

2. **A frozen reservoir hides the obstruction.** Upheld against exact inheritance. The old shell fails under a legal reservoir variation; no external coherent reservoir or fitted $\mu(\tau)$ was used.

3. **A sign change alone proves first order.** Upheld unless compactness and exclusion of small-charge minimizers are supplied. Both are proved above.

4. **Exchange must occur in the equilibrium minimizer.** Upheld against that interpretation. Generic nonequilibrium states exchange; static phases need not carry net exchange. No trajectory crossing, relaxation, seed generation or Gibbs equilibration is proved. Exact $\Psi = 0$ remains invariant.

5. **Finite statements supply physical time or a limit.** Upheld against promotion. K-dependent η_K, fixed spatial carrier and new canonical time remain explicit. The pairing need not have local gauge or full internal unitary invariance.

Primary and non-importing independent exact checks reproduce the rational window and exchange. Hostile review checked the alternative charge-norm exclusion estimate and the full reservoir variation. Six parameterized Lean lemmas cover cancellation, square completions and a negative trial, not the ODE, compactness or physical correspondence. External adversarial review of those analytic steps is invited. The original R-158 primary and independent certificates reran with 35 and 24 checks. Its old full verifier failed historical PDF byte determinism under the current engine; both changed PDFs were restored byte for byte. The old full verifier is not reported PASS and expected source hashes were not changed.

6 Result footer

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Result ID: R-578 (host C6-SPACETIME-SIGNATURE)
Precise statement: complete regulated charge-transfer flow;
controlled shifted variational first-order coexistence.
Exact old R-158 coexistence inheritance is rejected.
Scope: fixed T^3, each fixed finite K, new parent.
Dependencies: pinned P1/R158 identity and positivity;
inserted parent parameters and canonical bracket.
Evidence grade: ANALYTIC, EXACT, EXECUTED, CONDITIONAL.
Tier before / after: C6 T1 / T1; T-054 unchanged.
No-overclaim statement: no physical owner/time, Pre-A,
spacetime, QFT, gravity, continuum, mass gap or TOE.
Next required action: physical input, or a precise
error in the theorem; no automatic successor.
Reproduction command: python -X utf8
verification/scripts/exch001_verify.py
Expected output: PASS_SCOPED_PACKAGE and partial Lean.
Falsification gate: wrong symmetry, bounds or compactness.
```